

CONTEXT 1

MATTER IN THE

UNIVERSE

By the end of this chapter you will have covered the following material.

Science as a Human Endeavour

- Advances in science understanding in one field can influence other areas of science, technology and engineering ([ACSCH011](#) and [ACSCH050](#))
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations ([ACSCH012](#) and [ACSCH051](#))
- The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences ([ACSCH013](#) and [ACSCH052](#))
- Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions ([ACSCH014](#) and [ACSCH053](#))



Contextual story – Matter in the universe

- Where elements come from
- Manipulating and using elements
- Elemental isotopes

Science as a Human Endeavour (SHE)

- Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)
- The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)
- The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)
- Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions (ACSCH014 and ACSCH053)

Chemical understanding

- Atomic structure
- Elemental spectra and spectroscopy
- Periodic table
- Mass spectrometry
- Covalent bonding
- Metallic bonding
- Nanotechnology – elemental carbon
- Isotopes

Science Understanding (SU)

- Trends in the observable properties of elements are evident in periods and groups in the periodic table (ACSCH016)
- The structure of the periodic table is based on the electron configuration of atoms, and shows trends, including in atomic radii and valencies (ACSCH017)
- Atoms can be modelled as a nucleus surrounded by electrons in distinct energy levels, held together by electrostatic forces of attraction between the nucleus and electrons; atoms can be represented using electron shell diagrams (all electron shells or valence shell only) or electron charge clouds (ACSCH018)
- Flame tests and atomic absorption spectroscopy are analytical techniques that can be used to identify elements; these methods rely on electron transfer between atomic energy levels (ACSCH019)
- The properties of atoms, including their ability to form chemical bonds, are explained by the arrangement of electrons in the atom and in particular the stability of the valence electron shell (ACSCH020)
- Isotopes are atoms of an element with the same number of protons but different numbers of neutrons; different isotopes of elements are represented using atomic symbols (for example, $^{12}_6\text{C}$, $^{13}_6\text{C}$) (ACSCH021)
- Isotopes of an element have the same electron configuration and possess similar chemical properties but differing physical properties, including variations in nuclear stability (ACSCH022)
- Mass spectrometry involves the ionisation of substances and generates spectra which can be analysed to determine the isotopic composition of elements (ACSCH023)
- The relative atomic mass of an element is the ratio of the weighted average mass per atom of the naturally occurring form of the element to 1/12 the mass of an atom of carbon-12; relative atomic masses reflect the isotopic composition of the element (ACSCH024)
- Nanomaterials are substances that contain particles in the size range 1–100 nm and have specific properties relating to the size of these particles (ACSCH028)
- The characteristic properties of metals (for example, malleability, thermal conductivity, electrical conductivity) are explained by modelling metallic bonding as a regular arrangement of positive ions (cations) made stable by electrostatic forces of attraction between these ions and the electrons that are free to move within the structure (ACSCH032)
- Elemental carbon exists as a range of allotropes, including graphite, diamond and fullerenes, with significantly different structures and physical properties (ACSCH034)

Investigations

Context 1

- Activities 1.1, 1.2, 1.3, 1.4, 1.5 and 1.6
- #### Chapter 1
- Activity 1.1
 - Experiments 1.1 and 1.2

Science Inquiry Skills (SIS)

- Conduct investigations, including the use of devices to accurately measure temperature change and mass, safely, competently and methodically for the collection of valid and reliable data (ACSCH003)
- Represent data in meaningful and useful ways, including using appropriate graphic representations and correct units and symbols; organise and process data to identify trends, patterns and relationships; identify sources of random and systematic error and estimate their effect on measurement results; and select, synthesise and use evidence to make and justify conclusions (ACSCH004)
- Select, construct and use appropriate representations including chemical symbols and formulae, molecular structural formulae, physical and graphical models of structures, chemical equations and thermochemical equations, to communicate conceptual understanding, solve problems and make predictions (ACSCH006)
- Communicate to specific audiences and for specific purposes using appropriate language, nomenclature, genres and modes, including scientific reports (ACSCH008 and ACSCH047)

Where elements come from

The elements that you see every day, the metals and non-metals, in compounds and mixtures, all had their beginnings in stars. A star is a giant ball of very hot gases held together by gravity. Most stars, including our Sun (Figure 1.1), are mostly hydrogen and helium. A process called stellar nucleosynthesis causes heavier elements to form gradually in larger stars, until we get the elements that lead to the formation of planets such as our Earth. Before we consider the formation of elements, we need to go back further, to the very beginning of the universe, to find out how the basic atomic particles, and the first atoms, formed.

◀ **Figure 1.1**
This star emits light due to fusion of hydrogen and helium.



To revise atomic structure, refer to Chemistry section 1.1 on page 104.

The Big Bang

Around 13.77 billion years ago, the universe came into being during an event called the **Big Bang**. The current theory states that the universe expanded very rapidly from a very dense and hot state and is still slowly expanding. In the beginning, there were no particles, just a lot of energy. About 10^{-43} seconds after the universe came into being, the first elementary particles formed, and after about 3 minutes, the first protons, neutrons and electrons were formed.

Over the next half a million years, hydrogen and helium nuclei came into existence, and after this the first stars formed and started to undergo fusion. Figure 1.2 on page 4 shows a timeline of the major events of the Big Bang.

Star life cycles

Stars can be many different sizes and temperatures, but all stars undergo nuclear fusion for about 90% of their life. During nuclear fusion, hydrogen is converted into helium inside the core of the star, and a vast amount of energy is released. During this time, the star is referred to as a **main sequence** star, as seen in Figure 1.3 on page 5. At the end of their life, stars start to behave differently, depending on their initial size. Figure 1.3 shows the different possibilities for a star after all its hydrogen has been used up and it is no longer a main sequence star.

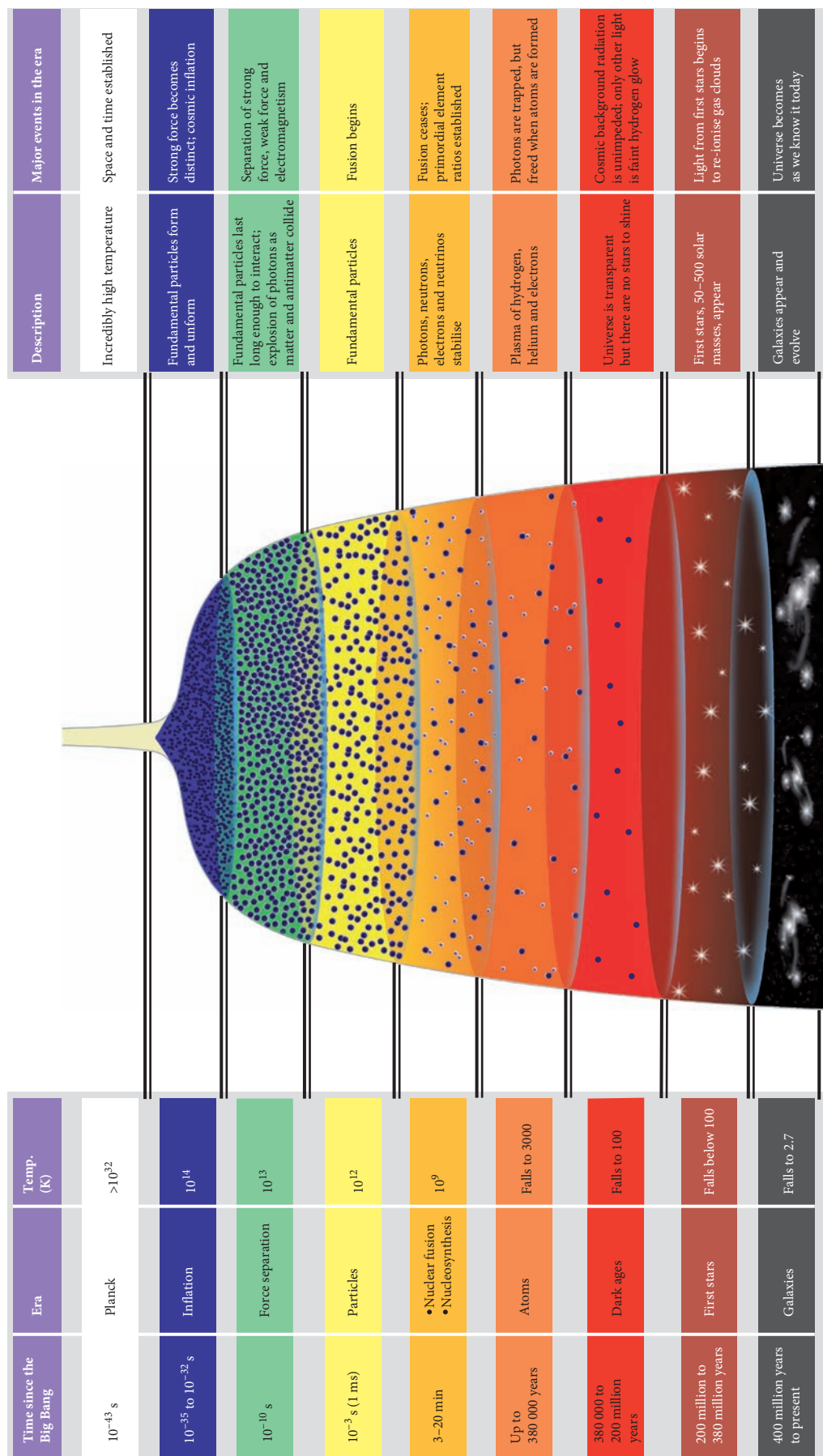
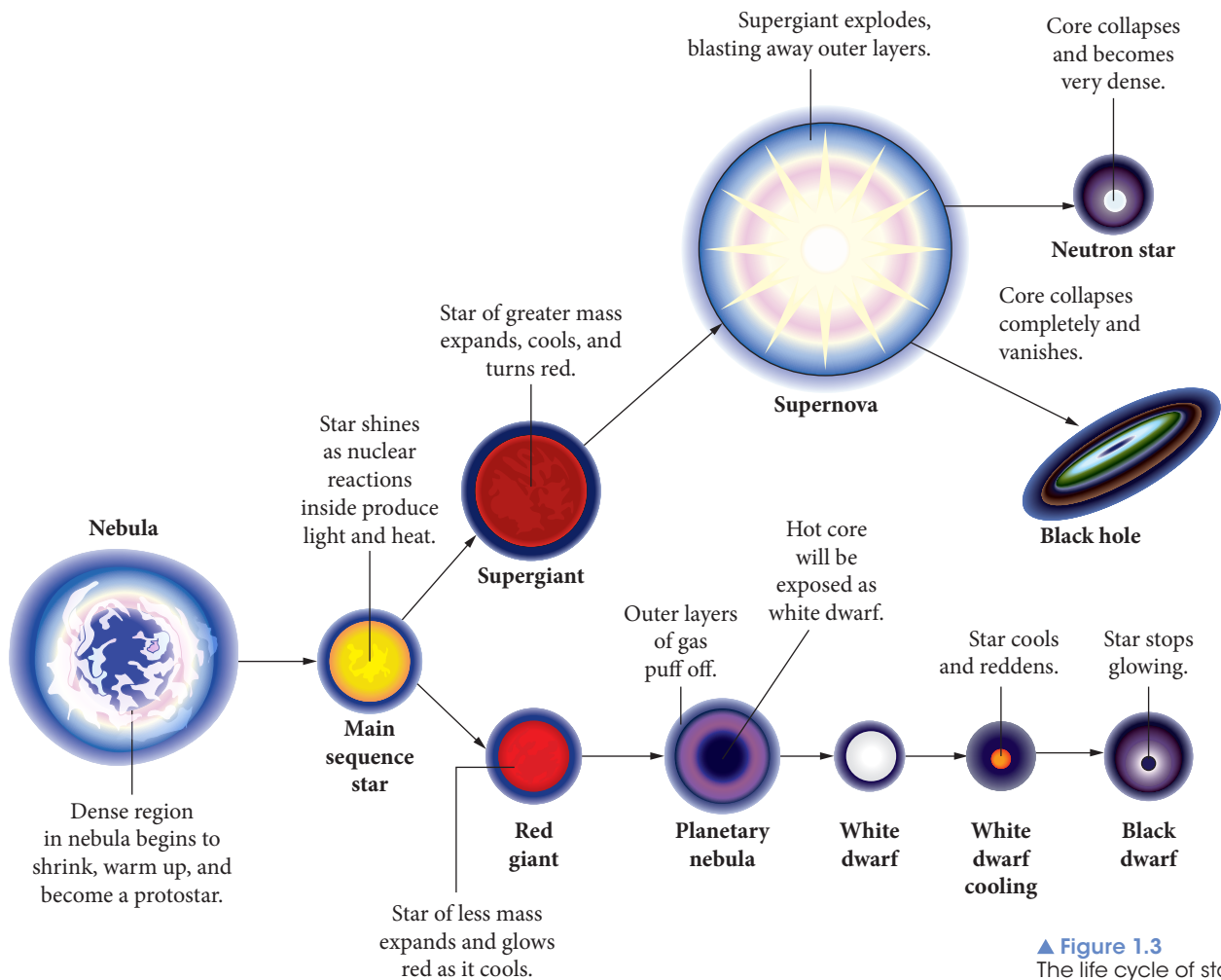


Figure 1.2 ▲ The Big Bang: timeline of major events



▲ **Figure 1.3**
The life cycle of stars

Our Sun is one of the smaller stars in the sky. When all its hydrogen has been used up, in billions of years' time, it will expand and cool, forming a **red giant**. Eventually, it will contract and form a **white dwarf**. Much larger stars form **red supergiants**, also cooler and much larger than the original star. Supergiants eventually explode violently in a huge explosion called a **supernova**, as seen in Figure 1.4. After this explosion, the star may form a dense, fast-spinning **neutron star** or a very dense **black hole**. These larger stars are where the heavier elements formed.

Stellar nucleosynthesis

The core of a massive star becomes extremely hot, much hotter than in smaller stars such as our Sun. The core temperature of larger stars allows the formation of elements heavier than helium. Once certain temperatures are reached, helium begins to fuse to form elements such as lithium, carbon, oxygen and neon. As the star gets denser and hotter, the carbon begins to fuse to form elements such as sulfur and silicon. Silicon fuses to form heavy elements such as scandium, chromium and iron. A massive star at the end of its life looks a little like an onion. It has layers of elements – the lighter elements are on the outside and the heaviest elements are at the core. You can see this in Figure 1.5 on page 6.

NASA, ESA, J. Hester and A. Loll, Arizona State University



▼ **Figure 1.4**
A supernova, the remains of a massive star

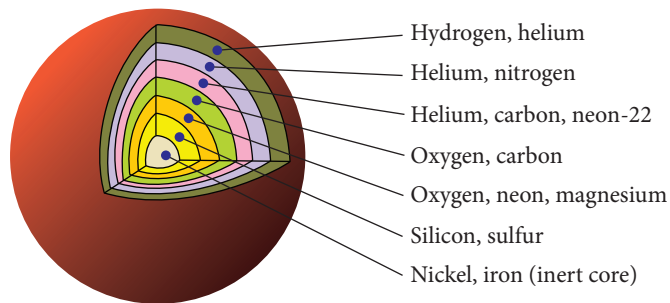


Figure 1.5 ▲
At the end of its life, a very large star has layers of elements.

After iron has formed, the star is unable to continue making heavier elements because the star is not hot enough to cause iron to undergo fusion. When the star reaches this point in its life cycle and it is not undergoing fusion, then it is not producing enough energy to be stable any more. The star will then explode in a supernova. The huge explosion creates a massive amount of energy, which allows the formation of the elements that are heavier than iron. The temperatures that are not possible inside a star exist inside the supernova. The heavier elements can now fuse. All the other natural elements in the periodic table are formed in the supernova explosion. So, without massive stars, the elements that we know and use on Earth today would not exist.

Analysing stars

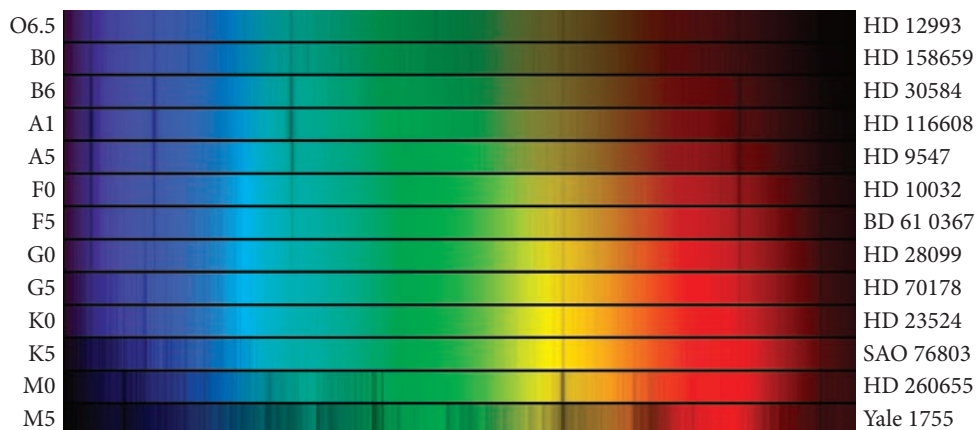
Even the stars within our solar system are too far away to travel to, so scientists must study them by looking at the light they emit. The most common instrument for analysing light is a **spectroscope**. This device breaks up visible light, and some non-visible light, into its component wavelengths to produce an **emission spectrum**.

Stars are made up of many elements. Each element emits a unique set of wavelengths, so an emission spectrum for a star consists of many lines. This is because the spectrum consists of a mixed pattern of all the elements together. Figure 1.6 shows the spectra of some stars – there are thousands of lines.

HOW DO ELEMENTS FORM IN STARS?

Watch this video to see how elements heavier than hydrogen and helium form in stars.

Scientific knowledge can enable scientists to offer valid explanations and make reliable predictions (ACSCH014 and ACSCH053)



Science Photo Library/National Optical Astronomy Observatories

Figure 1.6 ▲

Spectra of 13 stars, each of a different type. The types (from top to bottom) are O6.5, B0, B6, A1, A5, F0, F5, G0, G5, K0, K5, M0 and M5. Our Sun is a G2-type star. The order OBAFGKM ranks the stellar spectral types in order of decreasing surface temperature, while the numbers 0–9 provide subtypes. An O-type would be hotter than 25 000 Kelvin (K) and appear blue. An M-type would appear red and be cooler than 3500K. The dark spectral bands show stellar composition.

Scientists can make a pretty good guess at the most likely elements that are in a star. For example, hydrogen and helium are always present in a star, so scientists look for their emission lines at known wavelengths first. The remaining lines are then compared with known elemental spectra to determine what other elements are present in the stars. For example, if all the known lines for carbon are present, then carbon is present in the star.

This technique is how the element helium was discovered in 1868. Scientists examining the spectrum of the Sun found a line at a wavelength that did not match any known element. Astronomers Jules Janssen and Norman Lockyer independently discovered the line around the same time and are given credit for the discovery of helium. The new element was named helium after Helios, the Greek god of the sun. It was not until 1895 that helium was eventually discovered in Earth's atmosphere.



To learn why atoms produce unique line spectra, refer to Chemistry section 1.6 on page 113.



To revise elemental spectra, refer to Chemistry section 1.8 on page 123.

ACTIVITY 1.1

IDENTIFYING ELEMENTS IN STARS

Aim

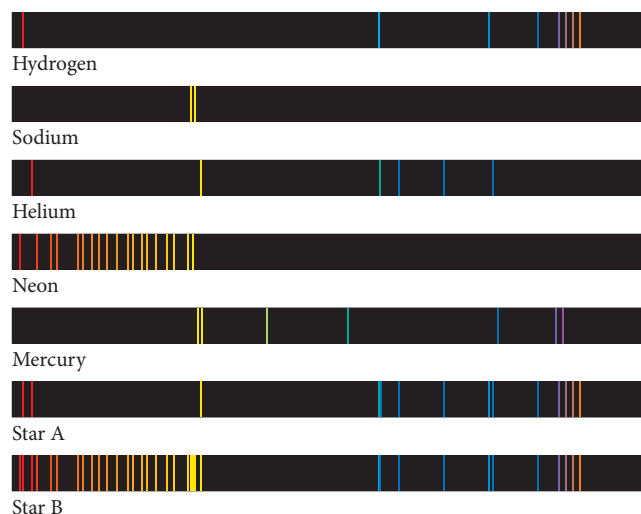
To identify the elements present in stars based on their emission spectra

What to do

Use your knowledge about line emission spectra to identify the elements present in star A and star B in Figure 1.7.

What did you discover?

- 1 Explain how you identified the elements in the two stars.
- 2 Which of the two stars is more likely to be a massive star? Why?



▲ Figure 1.7
Elemental and star spectra

Synthetic elements

There are 92 naturally occurring elements that originated in stars and supernovae. The rest of the elements in the periodic table do not exist in nature – they are synthetic. Scientists use a number of different methods to make **synthetic elements**.

A number of other elements have been synthesised by scientists. All of them are unstable because the forces in their nuclei are unbalanced, which causes the nucleus to break down over time. As they break down, the nuclei emit particles and electromagnetic radiation. This process is known as **radioactive decay**. Sometimes, the particles or radiation emitted during radioactive decay can be useful. These will be discussed further on page 22.

To make a new element from an existing element, you need to add extra protons. One way to do this is to bombard an atom with nuclear particles such as protons, helium nuclei, neutrons or heavier atomic nuclei. Some of the new material becomes part of the nucleus of the element, giving it extra protons and forming a new element. Californium, atomic number 98, was formed this way in 1950 – helium ions were fired at the element curium, atomic number 96. The two protons in the helium nucleus added to the curium nucleus, giving it 98 protons and forming a new element.

Organising elements

Scientists organise the elements in the form of the periodic table. Many elements have been known about since prehistoric times, but the scientific concept of elements wasn't devised until around the 17th century when phosphorus as an element was officially discovered. By 1900, there were about 80 recognised elements and the first attempts at organising a system of categorising them really started to take place.

The first attempts at organisation worked around a system proposed by German chemist Johann Dobereiner. In 1817, he noticed that elements that had similar properties could be grouped in groups of three, which he called triads. Dobereiner noted that the groups of three elements had similar properties and their atomic masses showed a specific pattern. Calcium, strontium and



COSMIC ORIGIN SPECTROGRAPH

Watch this video about the Cosmic Origin Spectrograph installed on the Hubble Space Telescope and see how it is used to find out about the universe.



To revise the periodic table, refer to Chemistry sections 1.4 and 1.7 on pages 111 and 119.

Figure 1.8 ►
Dobereiner's triads on
the current periodic
table

H							He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar
K	Ca	Ga	Ge	As	Se	Br	Kr
Rb	Sr	In	Sn	Sb	Te	I	Xe
Cs	Ba	Tl	Pb	Bi	Po	At	Rn

barium were one triad. They all have similar chemical properties and the atomic mass of strontium is almost exactly halfway between that of calcium and barium. Dobereiner noticed the same thing for lithium, sodium and potassium, so he called them another triad. Other triads were phosphorus, arsenic and antimony, and chlorine, bromine and iodine. Figure 1.8 shows where these groups of three elements appear on the current periodic table.

The modern periodic table was developed by Dmitri Mendeleev and published in 1869. Mendeleev wrote out the names of the elements with their atomic masses and other properties. He laid them out in rows (periods) and columns (groups) by first ordering the elements by mass. He then organised them so that elements with similar properties were in the same group. When he finished, there were gaps in his periodic table. These gaps were later filled as more elements were discovered that had properties and masses that fitted exactly into the gaps. This table has been used with only slight modifications since then. The table has recently been revised to include the newly discovered noble gases, the rare earth elements called the **lanthanoids** (also called lanthanides) and the radioactive **actinoids** (also called actinides). The rare earth elements do not fit in with the periodic nature of the rest of the table so are placed underneath it.

*Scientific knowledge
can enable
scientists to offer
valid explanations
and make reliable
predictions
(ACSCH014
and ACSCH053)*



ACTIVITY 1.2

BUILDING YOUR OWN PERIODIC TABLE

Aim

To construct a periodic table based on element properties and masses

You will need

- A set of element cards with information about the elements and their chemical properties, pre-made cards from the Internet, or an interactive periodic table to create your own

What to do

- If you are making your own cards using the interactive periodic table, decide what properties to include on each card.

Element properties could include:

- name and symbol
- appearance
- state
- atomic number
- atomic mass
- boiling point
- melting point
- valency.

- Arrange your element cards (either created or printed) in a line in order of increasing atomic mass.
- Identify where the properties repeat in the line and use this to create a logical set of rows that show trends.



ELEMENT CARDS

Use these sets of
element cards with
information about
the elements.

What did you discover?

- 1 Compare your table to the periodic table. Make a list of the similarities and differences between the two tables.
- 2 Examine the properties you looked at when constructing your table and explain why you had differences.
- 3 Rearrange your cards so they match the periodic table.
 - a Where are the metals in the periodic table? Look at the properties of the metals and list those properties that are common.
 - b Where are the non-metals in the periodic table? Look at the properties of the non-metals and list those in common.
 - c Identify any more trends in properties that you can see. Look down groups (columns) and across periods (rows).



INTERACTIVE PERIODIC TABLE

Use this interactive periodic table to see what elements look like and read their chemical information.

Manipulating and using elements

Most of the substances on Earth are composed of multiple elements in the form of compounds or mixtures, but there are still a lot of elements that are used or studied in their pure form. We analyse materials for particular elements, for commercial, health or forensic reasons. We use elements such as metals in a variety of fields, including construction. A newer application is the use of elements in nanotechnology to create materials with very specific applications.

Analysing elements

Element analysis can focus on identifying the element present – this is called **qualitative analysis**. But if you know which elements are present, then you can focus on measuring how much of the element is present. This is called **quantitative analysis**. A number of techniques, such as spectroscopy, involve the absorption and emission of light by atoms of elements as electrons move between energy levels.



To understand more about and revise spectroscopy and spectrometry, refer to Chemistry sections 1.8–1.10 on pages 123–131.

Heavy metal poisoning

Certain metals are extremely toxic to humans and if ingested in large enough amounts can cause severe illness, DNA damage and even death. Mercury, lead, aluminium, arsenic and cadmium are found in small amounts in humans because of everyday contact with materials and the environment. These elements are sometimes known as heavy metals because they have quite high atomic masses. Some of them can be seen in Figure 1.9.

Lead



Cadmium



Arsenic



▲ Figure 1.9

Some heavy metals: lead, cadmium and arsenic

Mercury is still used in amalgam dental fillings. People consume mercury-containing compounds in fish due to industrial pollutants in the ocean and the effects of biological magnification. This is where small animals consume mercury and a large number of these animals are eaten by larger fish and animals. The amount of mercury in the larger animals is magnified as they consume so many of the smaller animals. Mercury acts as a neurotoxin in humans and can affect development and functioning of the nervous system.

Mercury liquid



Science Photo Library/Cordelia Molloy

Mercury-containing amalgam fillings



Science Photo Library/CNRI

A mercury thermometer



Shutterstock.com/Andrii Gorulko

Figure 1.10 ▲

Mercury has various uses, including in dental fillings and thermometers, but it is toxic.

Lead is used in paints and pigments to give them useful properties. Recently, a large number of toys made in China were recalled. Children in the USA and Europe had become violently ill after playing with toys that had paints and pigments containing lead. Researchers from Greenpeace bought 500 toys in five Chinese cities and conducted tests that showed one in 10 of the toys contained dangerous levels of lead with one containing 1200 times the amount of lead permitted under European safety standards.



Shutterstock.com/Gillian Entwiss



Alamy/Robert W. Ginn

Figure 1.11 ▲

Lead-based substances

Accumulation of aluminium in the human body can lead to osteoporosis, headaches and problems with liver and kidney function. You come into contact with aluminium through deodorants, drinking water, medications and some processed foods.

Arsenic is a very toxic metal that is less commonly used than the metals discussed so far. It is probably most famous as a method of poison in crime novels. Small amounts of arsenic enter the human body through exposure around industrial sites and to arsenic-treated wood.

Cadmium is a metal that directly damages DNA in the body. Again, you are less likely to come into contact with cadmium but it is used to make batteries and in the mining industry.

Human tissue, blood and urine can be analysed for these toxic metals by a number of different spectroscopic methods. Spectroscopy is a useful analytical method because it generally doesn't destroy materials. Spectroscopy requires only very small amounts of the sample, such as hair, blood and urine, so that most of it can be saved for continued testing or for evidence in court cases.

Scientists have been trying to find the causes for developmental diseases such as autism. One theory is that ingestion of toxic metals could be a contributing cause. Recent studies have shown that children with autism often have significantly higher levels of toxic metals, such as mercury, lead and thallium, in their bodies. Mass spectrometry is used to distinguish heavy metals from elements you expect to find such as carbon, oxygen and hydrogen.

Traditional methods of analysis are often time-consuming and expensive. Spectroscopy allows scientists to quickly analyse information for a vast number of subjects. This increases the chances of finding the causes of problems.

Herbal medications are sold over the counter in a number of countries and are generally considered to be a healthy option for building up the immune system and helping with minor health issues. Regular testing is done on these medications to determine their content. Atomic absorption spectroscopy and X-ray fluorescence spectroscopy testing has found alarming levels of heavy metals, including lead, mercury and arsenic, in some herbal remedies. Sometimes these metals have been added deliberately. Alternative herbal medications popular in India and China combine herbs with metals, minerals and gems. These medications are then sold online all over the world where people have been warned to be aware of what they are taking. Some tests have shown metal levels up to 10 000 times higher than the limits set by health professionals for some tablets.

Forensic testing

Forensic science uses spectroscopy in many ways. A big problem with forensic science is that most analytical methods destroy the evidence. This means that the evidence cannot be subject to further testing and cannot be presented in court. Methods such as spectroscopy that do not destroy the sample are in increasing demand. New methods of analysis are being developed all the time.

American chemist Professor Kenneth Busch developed a method of examining skeletal bones to determine a more accurate time of death, or post-mortem interval. After a person dies, their bones start to lose water and the molecules inside start to decompose. Busch developed a process that uses diffuse reflectance near-infrared spectroscopy. He collected the patterns of spectra from decaying pig bones over a 3-month period and identified a trend as the bones aged. A mathematical model was developed that allowed the post-mortem interval to be determined when bones are analysed. An important feature of this technique is that it uses infrared radiation, which does not destroy the substance during the analysis.

▼ Figure 1.12

Due to the potential harmful effects of cadmium, some types of battery are now advertised as being free of this metal as well as mercury.



Alamy/Beaconstock

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)



Alamy/Adam Gault

▲ Figure 1.13

Professor Busch's process of diffuse reflectance near-infrared spectroscopy can determine the post-mortem interval of decaying bones such as this skeleton.



To learn about chromatography, refer to Chemistry section 6.9 on page 289.



To revise mass spectrometry, refer to Chemistry section 1.10 on page 131.

In legal cases, documents such as ransom demands, receipts, wills and threatening letters are often involved. Analysis of inks can provide evidence of someone's guilt or innocence. For a long time, the methods involved in analysis destroyed the document, which meant that there was no evidence to display in court and legal cases were often lost. Traditional methods such as **chromatography** require part of the document to be removed, and the more analyses carried out, the more of the document is permanently destroyed.

Scientists in the USA have developed a system that allows immediate analysis of the whole document by mass spectrometry. This is called the DART (direct analysis in real time) system. The DART method removes a microscopic amount of ink from the document, leaving the rest of the document intact. Visually, you would never know that anything was missing. The minute sample of ink is analysed, and a spectrum is produced, which is compared against a very large library of ink samples. Examples of the spectra produced by different inks are shown in Figure 1.14. This method is often used to show when two different pens have been used on a single document to alter a receipt or forge someone's signature.

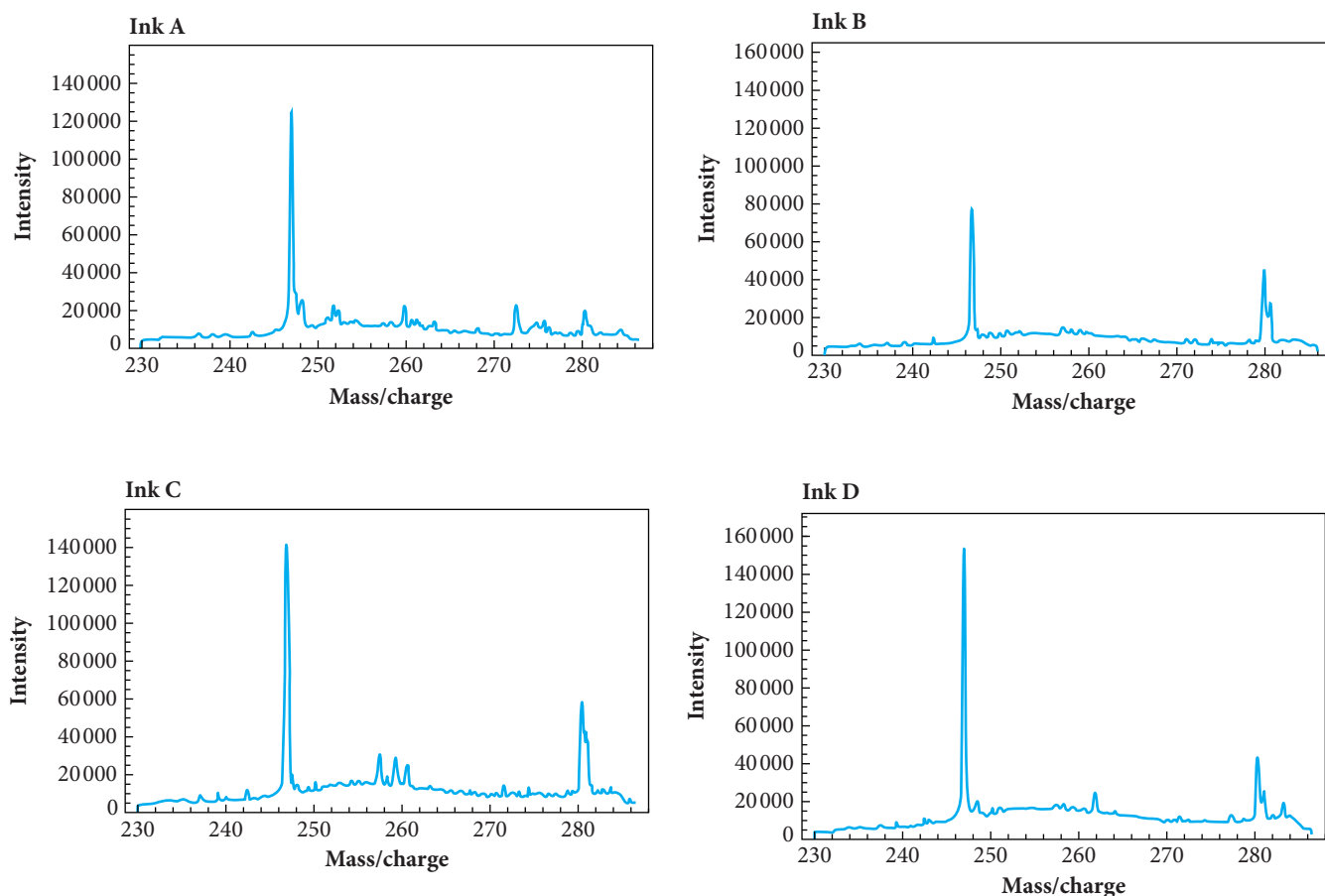


Figure 1.14 ▲

These mass spectra show the different composition of four different inks.

ACTIVITY 1.3

WHODUNIT?

Aim

To identify who forged their signature on a document by analysis of the ink spectra

You will need

- Figure 1.14

What to do

Read the following information carefully. Use Figure 1.14 to help you answer the questions below.

A handwritten will has been found and two siblings are fighting over the contents. The brother claims that he and his sister were each supposed to inherit half their parents' estate. However, this will states that everything is to be given to his sister. The brother suspects that his sister, who found the will, has changed part of it.

You are a forensic chemist assisting the police. You are given the note and the following information.

- Pens A and B were discovered in the parents' house.
- Pens A and C were discovered in the daughter's house.

You take samples of ink from the will and discover that most of the will is written in a pen that has the following characteristics in the spectra.

- There is a peak at a mass/charge value of approximately 247.
- This peak has an intensity value of approximately 80 000.

You also discover that the section of the will that names the daughter as the sole recipient is written in an ink that has the following characteristics in the spectra.

- There is a peak at a mass/charge value of approximately 247.
- This peak has an intensity of approximately 140 000.

What did you discover?

- 1 Briefly describe the process that you would have used as a forensic chemist to analyse the inks in the document.
- 2 Describe the information that can be determined from the spectral information given to you in this case.
- 3 Explain the chemical evidence that links the ink in the pen in the daughter's house to the changed will.

Food analysis

Many elements, such as calcium and iron, are necessary for good health. Other elements, such as heavy metals, can cause severe health problems at high enough concentrations, and should not be in food. Sometimes, food labels can be misleading or not specify all the ingredients contained in the food. Many analytical techniques are used to investigate claims and test foods for safety. The various types of spectroscopy play a major role in this analysis.

Certain foods such as cheese and wine are produced in various areas of the world. Countries and regions are very protective of their food products. For example, technically champagne is produced only in the Champagne region of France, but winemakers around the world have called their sparkling wines 'champagne'. This is no longer allowed and these champagne-style wines must be labelled as 'sparkling wines'. Wines labelled as 'champagne' are often tested to verify that they been produced in France.

The use of scientific knowledge is influenced by social, economic, cultural and ethical considerations (ACSCH012 and ACSCH051)



To revise how levels of contaminants are detected and measured, refer to Chemistry section 1.9 on page 127.

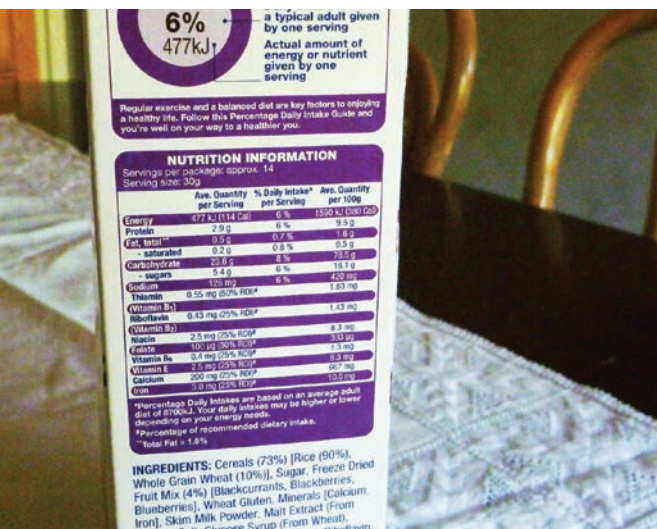


Figure 1.15 ▲
Composition of food as listed on the container

Cheese types can also be specific to certain areas; for example, true cheddar cheese comes from the region of Cheddar in the UK. In other cases, some cheeses include the milk of goats or sheep instead of cows, which can cause allergy problems. For this reason, it is important that cheeses are labelled accurately. Testing can determine whether companies are labelling their food products correctly.

There can be serious consequences when products are mislabelled. In some places, there have been problems with so-called beef mince containing other types of meat. It also can indicate substitution with cheaper grades of meat and possible health threats through contamination with untested or unregulated ingredients. People who practice certain religions are not allowed to consume pork. So if pork is present in food, but is not identified on the label, they inadvertently break religious laws. Regular testing of food detects contaminants.

ACTIVITY 1.4

FOOD RECALL

Aim

To devise a process for determining whether a food needs to be recalled due to metal contamination

What to do

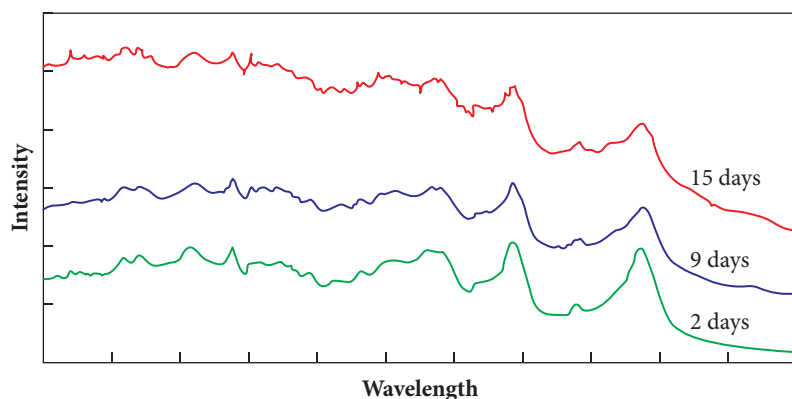
Read the following information carefully.

You work for the Food Standards Board in your state and you are given a sample of meat that is suspected of containing two types of metal particles. Under the food standards, mercury is allowable as long as it is below 0.5 parts per million. If cadmium is present in any amount, then the meat must be recalled and banned from sale.

In your group, devise a method for determining whether the food is contaminated past acceptable levels.

- 1 Using the information on spectroscopic techniques in Chemistry Chapter 1, explain how you would identify whether any cadmium or mercury was present.
- 2 Using the same information, explain how you would measure the concentration of mercury to determine whether it is below the recommended safe concentration.

In Germany, a project called FreshSCAN has developed an optical sensor that can work through food packaging to get instant feedback about the freshness of meat. No elaborate testing is required and no sample of food is needed for laboratory analysis. Scientists tested the spectroscopic properties of meat in varying stages of freshness and found that the spectra showed patterns that could then be used as a comparison for other samples. As meat ages, chemical changes affect how the atoms and molecules absorb and emit light. This leads to changes in the emitted spectra, as seen in Figure 1.16 on page 15. The benefits of this system are that feedback is instant and unsafe food can be removed immediately. The technology is being examined to see if it can be adapted to other foods by setting up a bank of spectra for foods of different levels of ripeness or spoilage.



▲ **Figure 1.16**

The spectra of meat change over time, which can be used to determine the age of meat on the supermarket shelf.

Soil pollution is a big problem. Water used for irrigation often contains industrial or agricultural pollutants, which are then transferred to the soil. As vegetables grow, they can take in these pollutants, which are then consumed by humans. By the time the food gets to supermarket shelves, it is too late. In France, a type of spectroscopy is being developed to obtain real-time data about contaminants in vegetables. This process is designed for use in the field. Traditional methods involve time-consuming laboratory work. This system uses a laser pulse to create a plasma sample of a very small amount of the vegetable while it is still growing. Plasma is a gas made up of electrons and ions, perfect for spectroscopic analysis. The plasma then undergoes spectroscopic analysis immediately in the field and the presence of specific metals can be detected. This system primarily analyses for toxic metals such as lead and cadmium, which cause health problems. So it is highly advantageous that food can be analysed before it gets to supermarket shelves and consumer's homes.



◀ **Figure 1.17**

Vegetables can be tested for dangerous levels of contaminants while they are still growing, before they appear on supermarket shelves.

Making useful materials

Almost everything around you is made up of two or more elements. Very few items you come into contact with are made from a pure element. The air you breathe contains elements such as oxygen and nitrogen, but when you breathe, you take in a mixture of air containing lots of different elements and compounds. Very few elements can be used in their pure form.



To revise electron arrangements in atoms, refer to Chemistry section 1.6 on page 113.



To learn about covalent bonding, refer to Chemistry sections 3.5 and 3.7 on pages 178 and 187.

Silicon

The element silicon is used a lot in technology. Very pure silicon (purity above 99.9%) is used primarily to create solar panels. Silicon is also used in **semiconductors**. A semiconductor is a material with an electrical conductivity somewhere between a conductor and an insulator. When a metal is heated, its conductivity decreases, but when a semiconductor is heated, its conductivity increases.

The silicon that is used in semiconductors has 'nine nines' purity, which means 99.999 999 9% purity. Silicon less pure than this will be inefficient as a semiconductor. Silicon is the best semiconductor because it can withstand the highest temperatures and electrical power running through it and still continue to function. This is important in computers and machinery because they heat up when they are used. A regular conductor will often fail when heat is applied but a semiconductor only gets better at conducting as it gets hotter. Silicon is such a good semiconductor because it has four electrons in its valence shell. When it bonds, it uses covalent bonding between atoms to form large lattice structures called crystals.

In crystal form, silicon is not highly conductive. The structure is slightly altered in a process called doping. This introduces different elements into the crystal structure. A common five-electron element in semiconductors is phosphorus. As seen in Figure 1.19, the phosphorus takes the place of one of the silicon atoms. An extra electron is present with nothing to bond to. This is called an n-type semiconductor.

If the silicon is doped with an element such as boron with only three electrons, then there is a space that is called an electron hole. Free electrons moving around the semiconductor will fill these holes. This p-type semiconductor can be seen in Figure 1.20.

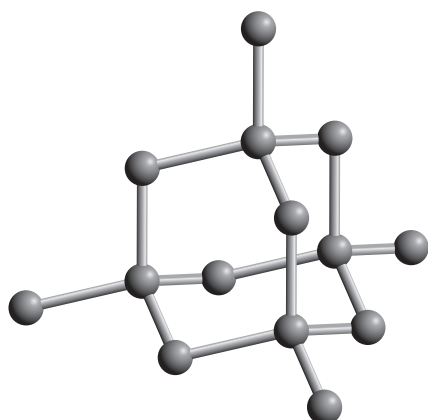


Figure 1.18 ▲
The crystal structure of pure silicon

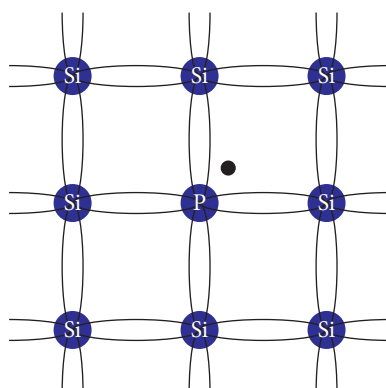


Figure 1.19 ▲
Phosphorus doping of silicon produces an n-type semiconductor.

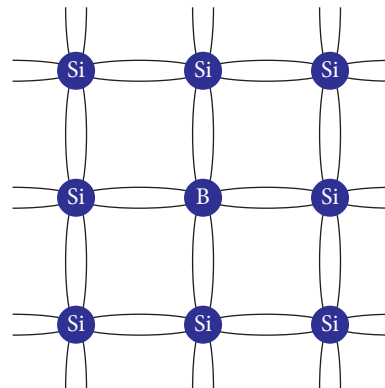


Figure 1.20 ▲
Boron doping of silicon produces a p-type semiconductor.



SILICON IN A SOLAR CELL

Use this interactive activity with animations and explanations to learn how silicon is used inside a solar cell.

When n-type and p-type silicon are put together, the extra electrons from the n-type move across to the p-type to fill the electron holes. The silicon crystal is now a moderate conductor of electricity and that is why it is called a semiconductor. The most common use for silicon in this form is in electronic components, such as transistors, and solar cells.

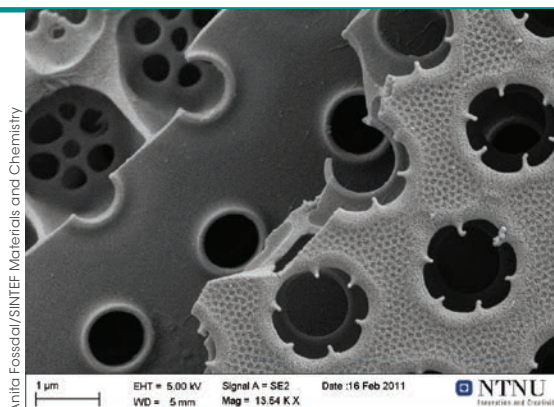
Using algae to generate electricity

Scientists are using single-celled algae called **diatoms** to improve the efficiency of solar panels. Specially shaped structures in their silica shells improve the diatom's use of sunlight in photosynthesis by about 30%.

Scientists grow billions of diatoms in tanks and coat them onto the surface of existing solar cells. The diatoms cover the surface in the most efficient manner for light capture and then overlay their silica shells onto the surface of the solar cell. The living material is removed, leaving a solar panel with increased efficiency. The silica shells that are left behind by the diatoms capture light in hexagonal holes and do not allow it to escape. Tests show that solar panels produced in this manner can triple the amount of electricity produced by traditional panels.

Figure 1.21 ▶

Pores in the silica shell of the diatom *Coscinodiscus walesii* completely trap sunlight for photosynthesis.



Titanium

The metal titanium has a number of useful properties in its elemental form. Titanium can be alloyed with other metals, such as iron, aluminium, vanadium and molybdenum, to form lightweight metals for use in the aerospace industry.

In its pure form, titanium has the highest strength-to-weight ratio of any metal. This makes it very useful in situations where metals of higher mass are not suitable. Titanium also has excellent corrosion resistance. It will not react with acids unless they are very concentrated. This means that in the environment, titanium will not react with the chemicals it comes into contact with. One of the most well-known uses for titanium is in medical implants as shown in Figure 1.22. It is referred to as a biocompatible material because it is non-toxic and is not rejected by the human body. Its lightweight and anticorrosive properties make it ideal for load-bearing structures such as ball-and-socket joints for hips and shoulders. Titanium pins, plates, bone screws, rods and staples are used throughout the human body to repair and set broken bones or even to replace body parts.



SOLAR PANELS FOR YOUR HOME

Watch this video to see how solar panels work and their advantages in terms of energy production.



To learn about metallic bonding, refer to Chemistry section 3.2 on page 163.



▲ Figure 1.22

A titanium hip joint a) prior to implantation and b) as seen on an X-ray



Alamy/Jain Cooper

Figure 1.23 ▲
Titanium jewellery is becoming popular because of its useful properties.

Titanium jewellery is gaining popularity because it is non-corrosive, lightweight and strong. Also, titanium jewellery is suitable for people with allergies who often react to other metals such as silver and gold.

Elemental gases

Most people are familiar with neon signs, but most of these signs do not involve the element neon. However, these signs all use elemental gases to create coloured light. Figure 1.24 shows a simple neon sign, while Figure 1.25 shows how neon signs are used in a city.

When elements are heated in their gaseous forms, they emit a specific colour of light related to the movement of electrons in the energy levels around the atoms. A 'neon' sign, regardless of what gas is used, works by applying an electric current to a tube of the gas. The current gives the electrons energy and a particular colour of light is emitted. Specific gases are selected to produce particular colours. Many signs incorporate different gas tubes to create a multicoloured effect. Table 1.1 shows the different colours that can be generated with different elements.



Shutterstock.com/FootToo

Figure 1.24 ▲
A simple neon light



Shutterstock.com/Lee Yiu Tung

Figure 1.25
Neon lighting in a city

Table 1.1 Neon sign colours and elements

Element	Colour
Neon	Red
Argon	Lavender-blue
Helium	Pale yellow
Sodium	Yellow
Krypton	Pale green-grey
Xenon	Blue-grey



To revise the emission of light by elements, refer to Chemistry section 1.8 on page 123.

ACTIVITY 1.5

USEFUL PURE METALS

Aim

To investigate the use of a pure metal and communicate this knowledge

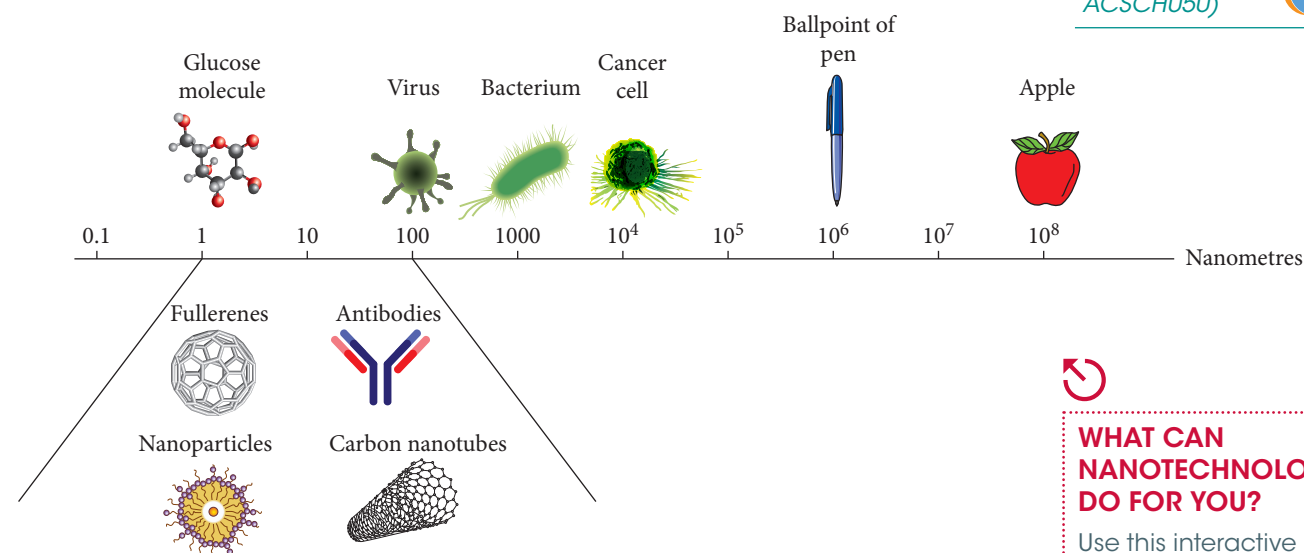
What to do

- 1 Research a metal that is used in its pure form.
- 2 Find out about the metal's uses and describe them.
- 3 Relate the uses of the metal to properties such as strength, hardness and conductivity.
- 4 Create a poster, PowerPoint presentation or short video to communicate your research.

Nanotechnology

The field of **nanotechnology** deals with matter at the atomic and molecular scale. **Nanomaterials** are materials that have particles between 1 and 100 nanometres (nm) in size. A nanometre is one-billionth of a metre. Objects this small cannot be seen with the naked eye or even by light microscopy. For example, a virus particle is around 40 nm while a single strand of DNA is around 2 nm. The diameter of an atom is even smaller than this, around 0.2 nm. For comparison, a single human hair is around 100 000 nm thick. So you can see that nanoparticles are extremely small.

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)



▲ Figure 1.26

The size of nanomaterials compared with different objects

Materials of this size have special properties that can be manipulated in order to perform specialised tasks. Applications of nanotechnology are already in use in medicine, materials fabrication, electronics and energy production. Two elements that have shown great promise in nanotechnology are carbon and gold. Carbon nanotubes are used in electronics and optics while gold is used in electron microscopy, electronics and materials science.

Carbon nanotubes

Carbon nanotubes are an allotrope of carbon. An allotrope is a different physical form of an element.

Carbon has four valence electrons so it will normally form four bonds in its compounds. Sometimes, carbon atoms bond to other carbon atoms but form only three bonds. When this happens, the carbons arrange themselves in a hexagonal, flat sheet that is known as



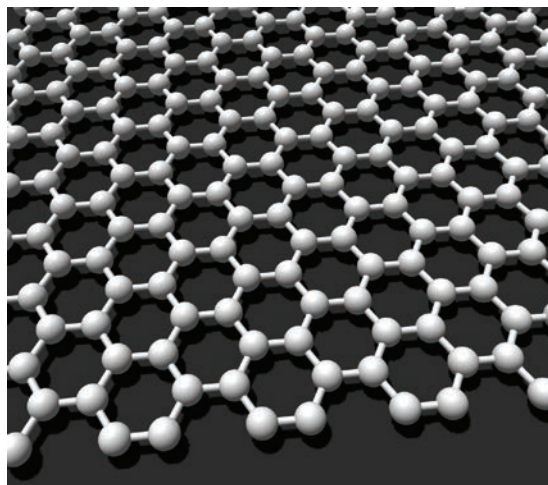
WHAT CAN NANOTECHNOLOGY DO FOR YOU?

Use this interactive activity to see how nanotechnology has applications in areas such as computers, medicine, sporting goods and textiles.



To understand more about carbon allotropes, refer to Chemistry sections 1.5 and 3.5 on pages 113 and 178.

Figure 1.27 ►
A flat sheet of graphene



graphene, as seen in Figure 1.27. Scientists use graphene sheets to form carbon nanotubes (Figure 1.28). Carbon nanotubes are very small, with a diameter as low as 1 nm. They can be as long as a few centimetres, although most are much smaller. In this form, carbon has the highest strength-to-weight ratio of any known material. When carbon nanotubes are constructed with multiple walls, as seen in Figure 1.29, their strength increases further still. Table 1.2 shows a number of projects that are using the strength of carbon nanotubes.

CARBON NANOTUBES

Watch this video to see how carbon nanotubes are used for a variety of applications.

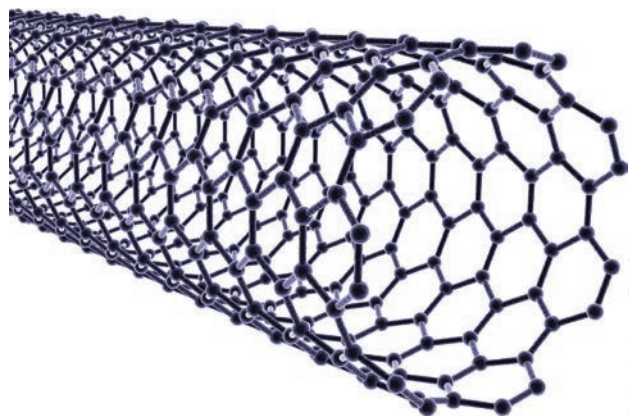


Figure 1.28 ▲
A carbon nanotube

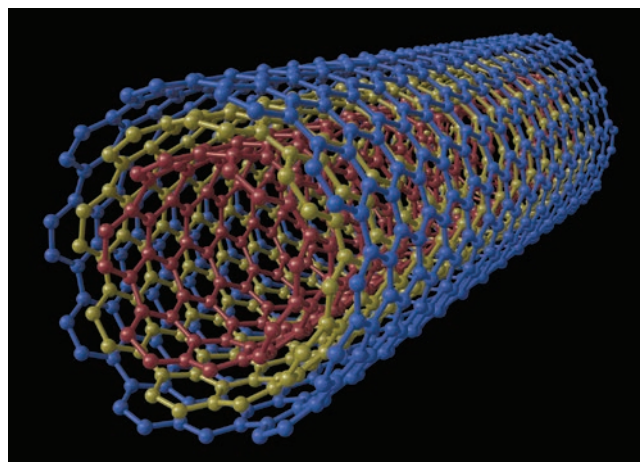


Figure 1.29 ▲
A multiwalled carbon nanotube

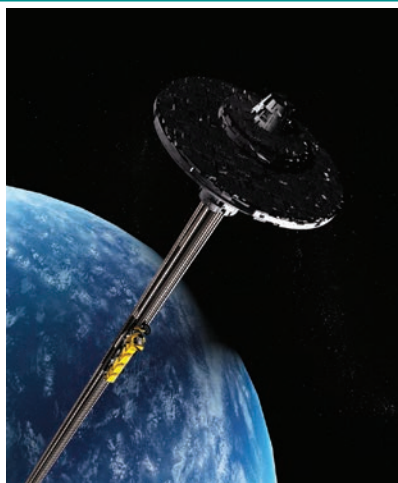
Table 1.2 Carbon nanotube projects

Area	Project details
Bulletproof armour	Lightweight body armour is being developed that allows the wearer more flexibility than current Kevlar materials. Nanotube armour is also stronger than Kevlar.
Synthetic muscles	Nanotubes are woven in with yarn and filled with wax. These muscles can lift weights 200 times heavier than natural muscle.
Setting broken bones	Nanotubes are used as a scaffold for new bone material to grow around.
Oil spill clean-up	Adding boron to nanotubes gives them sponge-like properties that allow them to absorb many times their weight in oil.
Morphing material	US National Aeronautics and Space Administration (NASA) has developed nanotubes that change shape when a voltage is applied. Controlling the voltage allows particular shapes to be formed.

Space elevator

Scientists at NASA are working to build a machine that allows objects to travel into space without having to be launched. It is proposed that this space elevator will lift materials into space at about one-fifth of the cost of launching by rocket. So far, no materials have been strong enough to construct the 35 000 km vertical column required to reach an orbiting space station.

Scientists from the University of Cambridge in the UK have combined short carbon nanotubes into long strands that could be the first steps towards a space elevator. The strands are flexible, but extremely strong and, most importantly, very light. All of these properties do not exist in natural materials, and it is only through the use of nanotechnology that a space elevator might one day become a reality. A space elevator could open the door to space tourism and colonisation of other worlds.

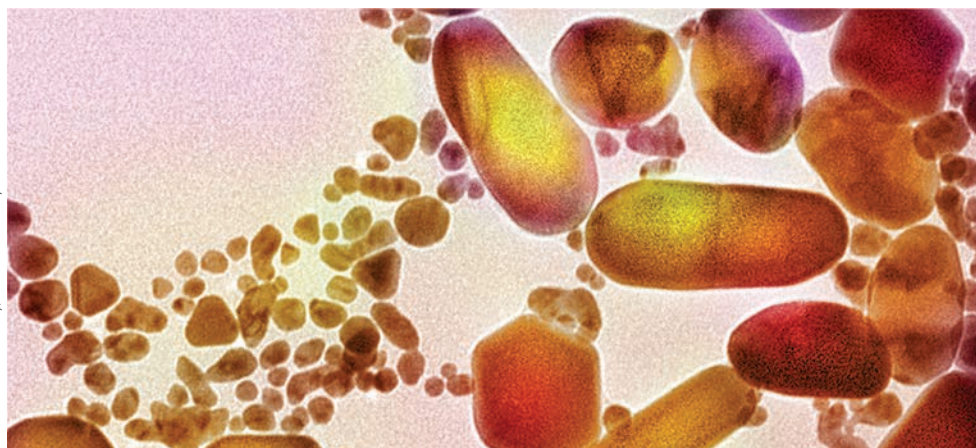


Science Photo Library/Christian Darkin

▲ **Figure 1.30**
An artist's impression of a carbon nanotube space elevator

Colloidal gold

Gold is a solid, heavy, dense metal used mostly for jewellery and decorative objects. But gold at a **nanoparticle** size looks and behaves very differently. Gold can exist as a **colloid** in a liquid, usually water, in which particles of gold about 100 nm in size are dispersed. Depending on the size of the particles, colloidal gold can be different colours. The most important property of colloidal gold is its ability to affect light in a predictable, measurable manner. Scientists can look at how light behaves around the particles and use this information in a number of ways.



Science Photo Library/David McCarthy

▲ **Figure 1.31**
A scanning electron micrograph of gold nanoparticles, used in cancer research and diagnosis

Gold nanoparticles have important applications in medicine. One of their most important uses is as markers. For example, rod-shaped gold nanoparticles called nanorods are being used in cancer diagnosis. The gold nanorods are synthesised so they can attach to the protein markers for a specific cancer, such as breast cancer. After the nanoparticles bind to the cancer cells, scientists examine how the protein–nanorod combination scatters light. This allows for very accurate information about the size, shape and invasiveness of the cancer. A more precise diagnosis leads to better treatment. This method is simpler and uses less expensive equipment than other methods of diagnosis.

The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)



EXPLORING NANOTECHNOLOGY THROUGH CONSUMER PRODUCTS

Watch this PowerPoint presentation on nanotechnology, and then explore various consumer products through the information provided.



To revise isotopes, refer to Chemistry sections 1.2 and 1.3 on pages 108 and 109.

Gold nanoparticles are also being used to fight cancer, not just detect it. The nanoparticles are synthesised to attach themselves to a particular type of cancer cell. After the gold particles are injected into the body and have attached themselves to the cancer cells, infrared lasers are shone through the body. The nanoparticles convert the otherwise harmless light into heat, which destroys the nanoparticle and any cell that it is attached to. The nanoparticles can also be used as a delivery system for drugs. The nanoparticles seek out the cancer cells, but are also attached to the chemotherapy drugs. These drugs target cancer cells much more effectively when bound to the cell surface with the nanoparticles.

Elemental isotopes

Most elements have at least two isotopes; some are stable and some are unstable. Both types of isotopes can be useful; however, unstable isotopes have particular properties that can be used and manipulated for a variety of purposes.

In an unstable isotope, the forces in the nucleus are unbalanced, so the isotope tends to emit particles and electromagnetic radiation from the nucleus. For example, carbon-12 and carbon-13 are stable and do not undergo radioactive decay. Carbon-14 is unstable and undergoes decay when a neutron in the nucleus converts into a proton. Another product of this conversion is an electron, which is emitted from the nucleus. This process is called beta decay. The electron emitted is known as a beta particle.

The **half-life** of a radioactive element is the time taken for half of the atoms in a sample to decay to a new element or isotope. The half-lives of these synthetic elements can vary from a few milliseconds to a year. The half-life of a particular element is constant and does not change. For example, an element may have a half-life of 1 minute. This means that after 1 minute, half of the atoms in a sample of this element would have undergone radioactive decay. One minute later, half of the remaining atoms would have decayed. After another minute, half of the remaining atoms would have decayed. If there were 100 atoms to start with, then after 1 minute there would be 50 atoms, after 2 minutes there would be 25 atoms and so on. A typical half-life graph is shown in Figure 1.32. In Activity 1.6 you will simulate the half-life of an element.

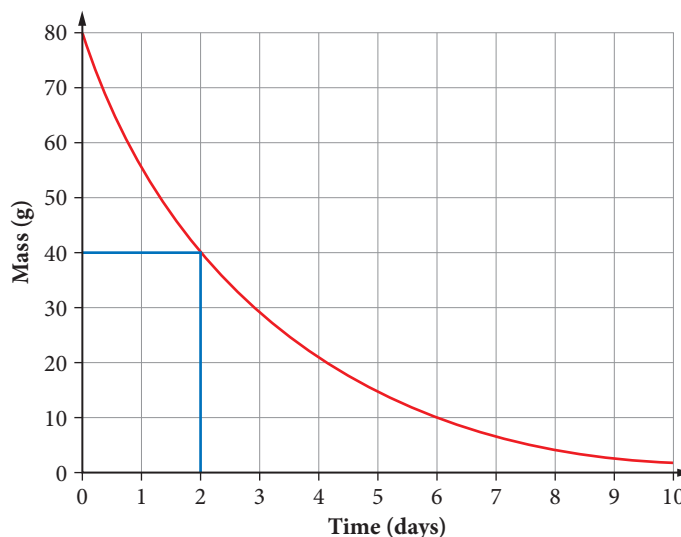


Figure 1.32 ▲

Graph of half-life for the radioactive decay of an element. The half-life is when half of the original sample has decayed, in this case 2 days.

ACTIVITY 1.6

DETERMINING HALF-LIFE OF AN ISOTOPE

Aim

To calculate the half-life of an element

You will need

- 100 M&M'S® or 100 small discs where one side can be easily distinguished from another
- Plastic bag

If you use M&M'S in this activity, perform it outside the laboratory.

What to do

- 1 Place the 100 M&M'S or discs into a bag and mix them thoroughly.
- 2 Pour out the M&M'S or discs over a clean piece of paper.
- 3 Remove the M&M'S that have the logo face down (or the discs with the blank face down). These have 'decayed'.
- 4 Count the M&M'S that have the logo face up (or the discs with the blank face up). These are still 'radioactive'. Record this number in a results table such as the one below.

Trial number	'Atoms' remaining	Combined class 'atoms' remaining
1		
2		
3		
4		

- 5 Return the remaining 'radioactive' M&M'S or discs to the bag and mix them thoroughly.
- 6 Repeat steps 3–6 until you have no more 'radioactive' M&M'S or discs.
- 7 Record the results of all the class on the board.

What did you discover?

- 1 On a graph, plot the trial number on the horizontal axis and the 'radioactive' atoms remaining on the vertical axis. Make sure you start your graph at zero. This is your half-life graph.
- 2 To find the half-life of your M&M'S or discs, draw a horizontal line from a point on the vertical axis that represents half the original number of M&M'S or discs to your graph line. Then draw a vertical line down to the horizontal axis. The place it cuts the horizontal axis is the half-life.
- 3 How many trials did it take for half the M&M'S or discs to 'decay'?
- 4 Did half of the M&M'S or discs decay in one trial? Suggest reasons why or why not.

Formation of isotopes

Some radioactive isotopes are made naturally. Carbon-14 is an example of this. When cosmic rays enter the atmosphere, they strike atoms and cause neutrons to be ejected. When a neutron collides with a nitrogen atom, a nuclear transformation occurs and carbon-14 is produced. The carbon-14 then enters the carbon cycle and takes part in normal processes such as photosynthesis, combustion and respiration. Carbon-14 isotopes behave chemically in the same way as carbon-12 because they have the same arrangement of electrons. When the arrangement of electrons is the same, the chemical properties of two substances are the same.

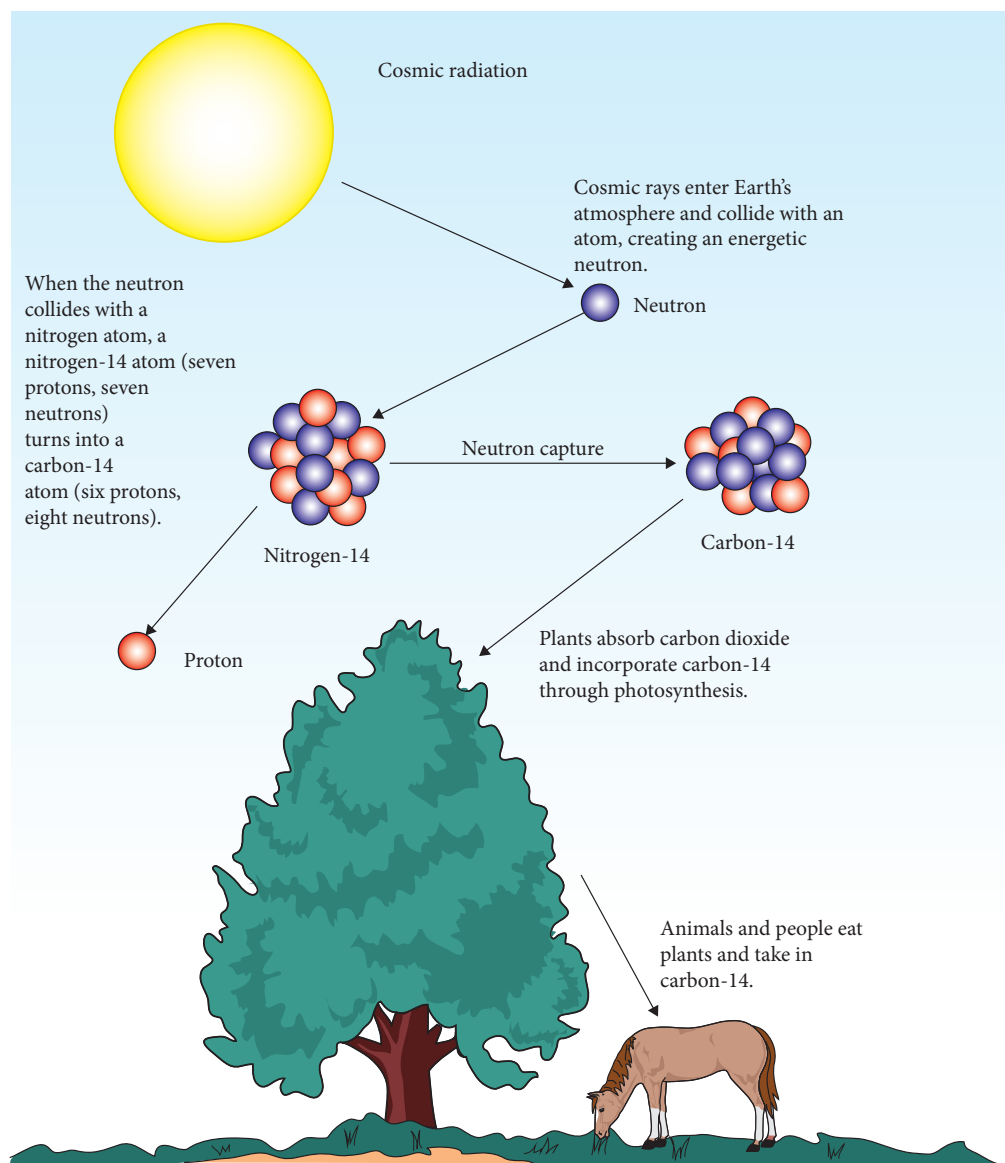


Figure 1.33 ►
Natural formation
of carbon-14 in the
atmosphere

Some radioactive isotopes that are formed only in stars do not ever make it to planets such as Earth because they are too short lived. For example, aluminium-26 is found in large amounts in stars, but has never been found naturally on Earth. About 3000 radioactive isotopes that are not found in nature have been made in nuclear reactors and particle accelerators. In a nuclear reactor, large elements undergo fission (splitting) to form smaller elements, which are usually unstable and radioactive. In a particle accelerator, small particles such as protons and electrons are fired at a nucleus to try to make it unstable either by causing it to accept extra particles into the nucleus or by knocking particles out. In both cases, the end result is an unstable nucleus that emits particles and radiation that can be detected and measured.

Uses of isotopes

Both stable and unstable isotopes are very useful in a wide variety of fields. The particles and radiation emitted by an unstable isotope can be detected to find the location of an isotope or measured to determine the quantity present.

Medical uses

Radioactive isotopes are used in medicine in two ways – to diagnose disease and to treat specific illnesses. Diagnosis of disease uses the radiation that is emitted from an unstable isotope. When an isotope is ingested by a patient, the radiation it emits forms a map of where it ends up in the body. The isotope can be used on its own or be attached to a molecule such as glucose that the body can use.

One of the most common radioactive isotopes used for diagnosis is technetium-99. This isotope has a short half-life of about 6 hours, which means technetium-99 breaks down very quickly and does not remain in the body. It is estimated that every year more than 20 million medical procedures use technetium-99. It has many uses, including imaging organs such as the kidneys, liver, lungs and brain. The isotope attaches to a specific pharmaceutical (a compound used as a medical drug) and is then transported to the required place in the body, where it decays, emitting radiation that is detected and measured. This is usually analysed by computer to produce images to help doctors diagnose their patients, as seen in Figure 1.34.

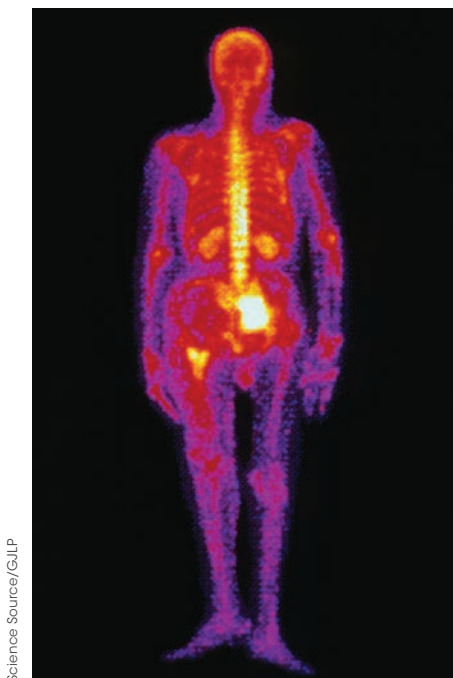
Radioactive isotopes are used in the treatment of diseases, especially cancer. Isotopes that emit radiation are ingested as a drink or in tablets. Cancer cells divide very rapidly so they absorb the radioisotope at high levels. The radiation emitted by the isotope as it decays damages the cells. The most common isotope used in this way is iodine-131. When this isotope is ingested, it concentrates almost exclusively in the thyroid so is used to treat thyroid cancer – the radiation emitted will target cancer cells in that area.

Mineral nutrition

The human body is made up of mostly organic materials containing carbon, hydrogen and oxygen. The presence of other elements such as calcium, sodium, zinc and iron in different forms is important to keep your body functioning. Some people, especially children, have problems absorbing minerals from the food. To diagnose this problem, isotopes are used to track how minerals move through the human body.

When the target group is children, pregnant women or older adults, radioactive isotopes are not recommended because the radiation that is emitted is particularly harmful for these people. Instead, enriched stable isotopes, which do not emit radiation, are used.

Calcium is an essential element for bone and teeth formation. The most common isotope of calcium is calcium-40. The heavier isotope calcium-42 is also usually present in very small amounts in any sample of calcium. Calcium-42 is not radioactive. Sometimes, children do not absorb calcium properly, so doctors measure the uptake of calcium by administering a sample of calcium enriched with calcium-42, which will not harm the child, but is detectable. Doctors measure the amount of the isotope that passes through the child's body and then calculate the amount of calcium absorbed. This allows doctors to identify whether that child has a problem absorbing minerals.



Science Source/Getty Images

Advances in science understanding in one field can influence other areas of science, technology and engineering (ACSCH011 and ACSCH050)



The use of scientific knowledge may have beneficial and/or harmful and/or unintended consequences (ACSCH013 and ACSCH052)



ALL ABOUT ISOTOPES

Watch this video from the Australian Nuclear Science and Technology Organisation, which explains how isotopes are formed and used.

◀ Figure 1.34

This image, produced by the radiation emitted by technetium-99, shows metastasised thyroid cancer.



Figure 1.35 ▲
People who work with radiation protect themselves against radioactive emissions by wearing gloves, masks and protective suits.

Tagging chemicals in reactions

Chemical reactions inside the human body can be examined by placing isotopic tracers into molecules. For example, a hydrogen-1 atom in a water molecule can be replaced with a hydrogen-2 atom. Hydrogen-2 is known as deuterium or heavy hydrogen. The heavy water molecule behaves in exactly the same way as other water molecules so it is used by the human body in the usual manner. This can then be tracked through the human body and the chemical reactions that occur can be investigated.

Chemical reactions in industry or in a laboratory can be studied by the same methods. Many chemical reactions have been known about for years and even

used in industry, but not understood properly. When a specific atom in a reactant is tagged with an isotope, the path that atom takes during a chemical reaction can be followed and recorded. This helps chemists understand the mechanisms of a reaction. When a reaction is understood properly, it can be manipulated. This allows chemists to make more of a particular product, or manipulate a reaction to get a product that may not otherwise form as readily. An example of this is tracking particular proteins during protein synthesis to produce artificial proteins that can then meet specific requirements of the chemist.

Safety of isotopes

Most radioactive isotopes do not pose significant risk because the particles and radiation they emit either are present in very small amounts or do not travel very far in air. Most people would not normally come into contact with isotopes that emit significant levels of dangerous radiation, apart from when undergoing certain medical procedures. People such as medical professionals, who work around radioactive isotopes, take appropriate precautions to limit their exposure through protective materials and secure storage. There are also restrictions on the number of procedures they can carry out each year, which further limits the amount of radiation they are exposed to.

The risk of damage from radioactive isotopes increases with the length of exposure. Radioactive isotopes that are used on or around humans normally have very short half-lives. This means that the total amount of radiation that humans are exposed to is very low as the isotopes decay to minimal amounts very quickly.

Table 1.3 Half-lives of some common radioactive isotopes used in medicine

Isotope	Use	Half-life
Technetium-99	Medical imaging of internal structures	6.01 hours
Iodine-131	Treatment of thyroid problems	8.02 days
Thallium-201	Testing for stress in patients with heart problems	3.04 days
Fluorine-18	Detection of cancer, through attachment to glucose molecules	109.8 minutes

CHAPTER SUMMARY

- Elements up to iron in the periodic table are formed during the life cycles of massive stars. Heavier elements are formed when stars die and explode in a supernova. Scientists can synthesise elements not found naturally on Earth.
- Light from the stars and techniques such as spectroscopy are used to learn about stars. Elements have unique emission spectra, so scientists can determine the elemental composition of a star by examining the light that it emits.
- Scientists organise elements on the periodic table to show patterns and trends. The periodic table was developed after scientists noticed patterns in the physical and chemical properties of elements.
- Scientists analyse materials by different techniques to identify elements present and to measure the quantities of those elements. Analysis of elements can provide information about health problems and the quality of our food, and help solve crimes.
- Elements in their pure form are useful in a wide variety of applications from electronics to medicine. Elements such as silicon, titanium and some elemental gases are used in semiconductors, solar panels, structural materials and signage.
- Nanotechnology deals with the behaviour of materials that has very small particles. Carbon nanotubes have extremely high strength-to-weight properties, which makes them useful in medicine, textiles and materials science. Colloidal gold is used widely in medicine because it can attach to specific cells and be targeted by light sources.
- Isotopes, both stable and unstable, are useful because of their differences from other isotopes of the same element. Isotopes form naturally and artificially when the nucleus of an existing element is changed through fission.
- Radioactive isotopes emit radiation that can be detected and measured. This radiation is used to diagnose and treat illnesses such as cancer. Risk of damage due to isotope exposure is minimised through strategies such as reducing exposure and using isotopes with short half-lives. Stable isotopes that do not emit radiation are used in some procedures involving high-risk groups.

CHAPTER GLOSSARY

actinoids the period of the periodic table that, with the lanthanoids, make up the f block

Big Bang the rapid expansion of matter from a high temperature and density state at the origin of the universe

black hole a region of space with an extremely high gravitational field from which light and matter cannot escape

chromatography techniques used to separate components of aqueous, liquid or gaseous mixtures

colloid a mixture in which tiny clusters of particles are dispersed through another substance; colloids do not settle out due to gravity and the particles are too small to be filtered

diatoms single-celled algae found in water sources

emission spectrum a pattern of bands produced by the emission of light from a source, separated due to the different wavelengths present

half-life the time taken for the radioactive emissions of an isotope to fall to half the original value

lanthanoids the period of the periodic table that, with the actinoids, make up the f block

main sequence stars that are in a state of equilibrium between gravity and the pressure produced by nuclear fusion

nanomaterial a substance that is made of or incorporate nanosized particles, with unique properties

nanoparticle a particle that is on the nanometre (one-billionth of a metre) scale (10^{-9} m)

nanotechnology a branch of science dealing with particles in the range of 1–100 nm

neutron star a celestial body composed of neutrons, with high density and strong magnetic field, formed after the death of a massive star

qualitative analysis analysis that identifies an element or substance but does not make numerical measurements about it, such as mass or concentration

quantitative analysis analysis that measures values such as amount, concentration or volume rather than just identifying the substance

radioactive decay spontaneous disintegration of an atom due to instability in the nucleus, releasing particles or electromagnetic radiation

red giant a star in the late stage of its life cycle, not undergoing hydrogen fusion, with high luminosity but low temperature

red supergiant a massive star that is undergoing helium fusion, with low temperatures and extremely large radii

semiconductor a substance that has little to no electrical resistance when cooled to extremely low temperatures

spectroscope a device used to separate light into its component wavelengths

supernova the explosion of a star at the end of its life cycle, resulting in the ejection of most of the star's mass

synthetic element an element that does not exist in nature, but has been made in a laboratory

white dwarf a planet-sized, dense star formed when a small star ceases nuclear fusion